

Capital University of Science and Technology

Department of Computer Science

Web Application Development

Semester Project Report

UnemployedNot

Job Portal Website

Submitted by:

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1. Introduction

UnemployedNot is a job portal website built for this semester's Web Application Development project. The idea behind the project is simple: give job seekers a single place where they can browse available job listings, apply to the ones that interest them, and keep track of the applications they have already submitted.

The project started as a static front-end assignment in earlier submissions and has now been converted into a fully dynamic web application. It is connected to a real database through a custom-built Web API, supports user registration and login, and allows complete CRUD operations on job applications. Users can sign up, log in, browse jobs, apply with a cover letter, and later view, edit, or withdraw their applications, all of which are stored and retrieved from the database in real time.

This report documents the technologies used, the system design, the features implemented, the structure of the database, and the major challenges faced while building the project along with how they were solved.

2. Technologies Used

The project uses a typical separation between frontend and backend, with the following technologies:

- Angular 19 - used for the frontend, built with standalone components and the newer `@if / @for` control flow syntax.
- ASP.NET Web API - used for the backend, handling authentication, job data, and application data.
- SQLite - lightweight database used to store users, jobs, and applications. The database file is created automatically using EF Core's `EnsureCreated()` method.
- Entity Framework Core (EF Core) - used as the ORM layer between the backend and the SQLite database.
- JWT (JSON Web Tokens) - used for authentication. Tokens are issued on login and expire after 7 days.
- Bootstrap 5.3 - used via CDN for layout and responsive styling.
- jQuery 3.7.1 - included via CDN for some smaller UI interactions.

3. System Architecture

The application follows a standard client-server architecture with the frontend and backend running as two separate projects communicating over HTTP.

3.1 Frontend

The Angular application runs on port 4200 and is responsible for everything the user sees and interacts with: the job listings page, the login and signup forms, and the My Applications page. It communicates with the backend purely through HTTP requests using Angular's HttpClient.

3.2 Backend

The ASP.NET Web API project runs on port 5039 and exposes REST endpoints for authentication, job listings, and job applications. It talks to the SQLite database through EF Core and returns JSON responses to the Angular frontend.

3.3 Authentication Flow

When a user registers or logs in, the backend validates the request and, on success, generates a JWT token containing the user's identity. This token is sent back to the frontend and saved. From that point on, an HTTP interceptor automatically attaches the token as an Authorization header on every outgoing request, so the backend can identify which user is making the request without the user having to log in again on every page.

4. Features Implemented

- User registration and login system, connected to the database, with form validation on both forms.
- JWT-based authentication, with the token automatically attached to all API requests via an HTTP interceptor.
- Reactive navbar that updates immediately after login or logout using an Angular signal, without needing a page refresh.
- Job listings page displaying available jobs.
- Apply for a job with a cover letter, which creates a new application record in the database.
- My Applications page where a logged-in user can view all applications they have submitted.
- Edit an existing application's cover letter directly from the My Applications page.
- Delete or withdraw an application, with a confirmation prompt before deleting.
- Empty state message with a "Browse Jobs" link shown when a user has no applications yet.

5. Database Design

The database consists of three main tables: Users, Jobs, and Applications. Each is described below.

5.1 Users Table

Field	Type	Notes
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Id	int	Primary key
Name	string	Full name
Email	string	Used for login
Password	string	Hashed password

5.2 Jobs Table

Field	Type	Notes
Id	int	Primary key
Title	string	Job title
Company	string	Hiring company

5.3 Applications Table

Field	Type	Notes
Id	int	Primary key
JobId	int	No foreign key constraint
JobTitle	string	Stored directly
Company	string	Stored directly
UserId	string	From JWT token
CoverLetter	string	Editable by user
Status	string	Application status
AppliedAt	DateTime	Timestamp

Note on foreign keys: the Applications table does not use a foreign key relationship to the Jobs table. Instead of storing only a JobId and looking up the job details through a navigation property, the JobTitle and Company fields are stored directly on the Applications table itself. This was a deliberate design decision made to solve a foreign key constraint problem encountered during development, explained in more detail in the Challenges and Solutions section below.

6. Challenges and Solutions

6.1 Foreign Key Constraint Error on Applications

Initially, the Application model only stored a JobId and used a navigation property to link back to the Jobs table. This caused a foreign key constraint error whenever a user tried to apply for one of the hardcoded jobs, because those jobs did not actually exist as rows in the Jobs table in the database.

Solution: the navigation property and foreign key relationship were removed entirely. The Application model was changed to store JobTitle and Company directly as plain string fields, alongside JobId as a plain int with no FK constraint. This meant the ApplicationController no longer needed any .Include() calls, and applying for a job worked without requiring that job to exist as a database row first.

6.2 Navbar Not Updating After Login

After a successful login, the token was being saved correctly, but the navbar still displayed the logged-out state (Login / Signup links) until the page was manually refreshed.

Solution: the AuthService was updated to use an Angular signal, loggedIn = signal(...), instead of a plain boolean property. The saveToken() method calls loggedIn.set(true) and logout() calls loggedIn.set(false). Because the navbar template reads this signal, any component referencing it updates automatically and reactively the moment the value changes, without needing a manual refresh.

6.3 My Applications Page Redirecting to Home

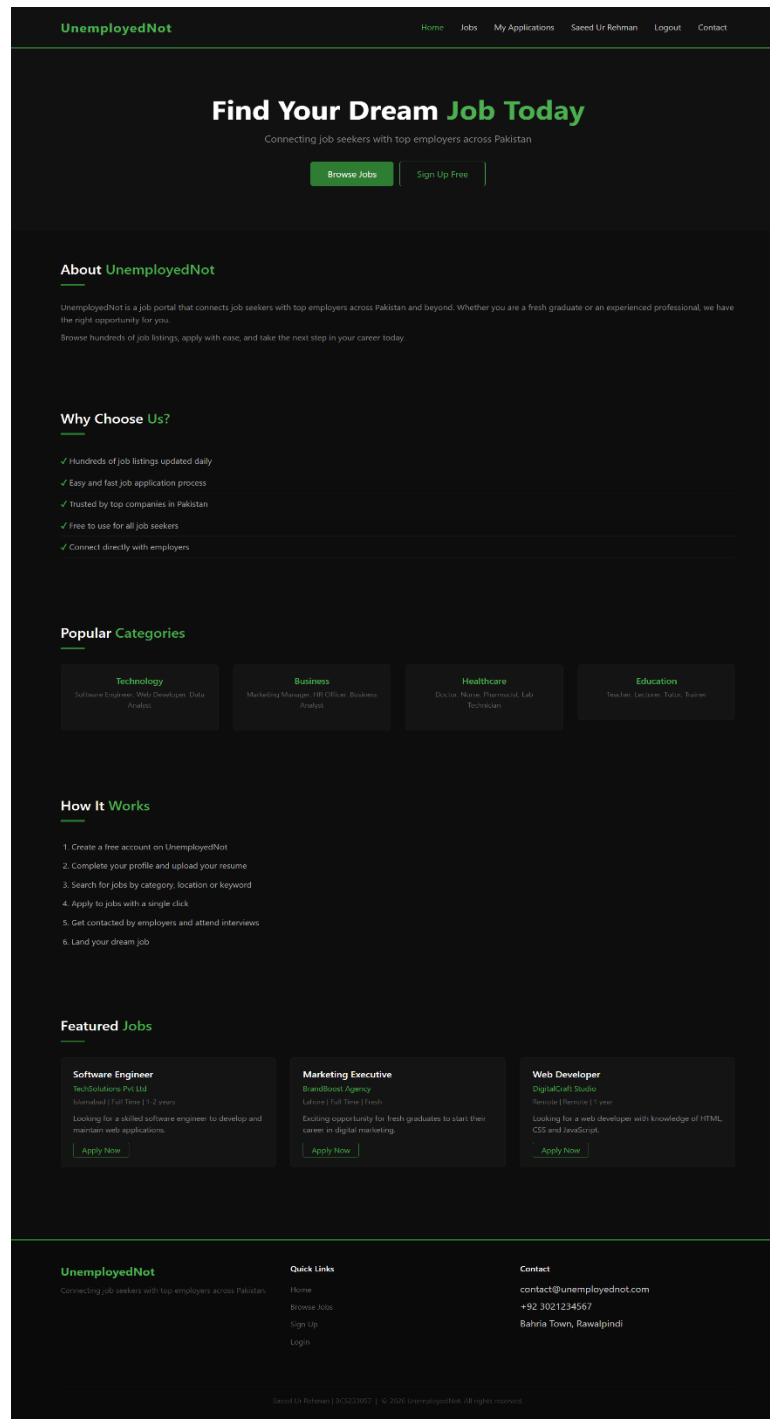
The My Applications page would briefly load and then redirect back to the home page instead of showing the user's applications.

Solution: two changes fixed this. First, the applications route was changed to use lazy loading instead of being eagerly loaded with the rest of the app, which resolved a timing issue with route guards. Second, an HTTP interceptor (auth.interceptor.ts) was added to automatically attach the JWT token to every outgoing request. Previously, requests to fetch applications were going out without the Authorization header, causing the backend to reject them and the frontend to redirect away. With the interceptor in place, every request is authenticated correctly and the page loads as expected.

7. Screenshots

The following screenshots demonstrate the working application. Each placeholder below should be replaced with the corresponding screenshot before final submission.

7.1 Home Page



7.2 Jobs Listing Page

UnemployedNot

[Home](#)
[About](#)
[My Applications](#)
[Search & Filters](#)
[Support](#)
[Contact](#)

Browse All Jobs

Find the latest job openings from top companies across Pakistan

Technology

Software Engineer

Software Engineer

Job Title: Software Engineer

Company: ABC Software Solutions

Location: Karachi, Pakistan

Apply Now

Web Developer

Web Developer

Job Title: Web Developer

Company: XYZ Web Solutions

Location: Lahore, Pakistan

Apply Now

Data Analyst

Data Analyst

Job Title: Data Analyst

Company: DEF Data Solutions

Location: Islamabad, Pakistan

Apply Now

Engineering

Civil Engineer

Civil Engineer

Job Title: Civil Engineer

Company: GHI Engineering

Location: Rawalpindi, Pakistan

Apply Now

Electrical Engineer

Electrical Engineer

Job Title: Electrical Engineer

Company: JKL Electrical Solutions

Location: Faisalabad, Pakistan

Apply Now

Mechanical Engineer

Mechanical Engineer

Job Title: Mechanical Engineer

Company: MNO Mechanical Solutions

Location: Multan, Pakistan

Apply Now

Medical

Medical Officer

Medical Officer

Job Title: Medical Officer

Company: PQR Medical Services

Location: Sialkot, Pakistan

Apply Now

Pharmacist

Pharmacist

Job Title: Pharmacist

Company: RST Pharmacy Solutions

Location: Gujranwala, Pakistan

Apply Now

Medical Lab Technician

Medical Lab Technician

Job Title: Medical Lab Technician

Company: UVW Medical Labs

Location: Bahawalpur, Pakistan

Apply Now

Education

Math Teacher

Math Teacher

Job Title: Math Teacher

Company: XYZ Education Services

Location: Dera Ghokhi, Pakistan

Apply Now

University Lecturer - CS

University Lecturer - CS

Job Title: University Lecturer - CS

Company: ABC University

Location: Peshawar, Pakistan

Apply Now

Business

Accountant

Accountant

Job Title: Accountant

Company: DEF Accounting Solutions

Location: Hyderabad, Pakistan

Apply Now

Bank Officer

Bank Officer

Job Title: Bank Officer

Company: GHI Bank Solutions

Location: Sukkur, Pakistan

Apply Now

Media

Content Writer

Content Writer

Job Title: Content Writer

Company: JKL Content Solutions

Location: Thatta, Pakistan

Apply Now

Digital Marketing Executive

Digital Marketing Executive

Job Title: Digital Marketing Executive

Company: MNO Digital Solutions

Location: Jamshoro, Pakistan

Apply Now

SEO Specialist

SEO Specialist

Job Title: SEO Specialist

Company: PQR SEO Solutions

Location: Sukkur, Pakistan

Apply Now

UnemployedNot

Find the latest job openings from top companies across Pakistan

Quick Links

- Home
- About Us
- My Applications
- Search & Filters
- Support
- Contact

Contact

For more information, please contact us at:

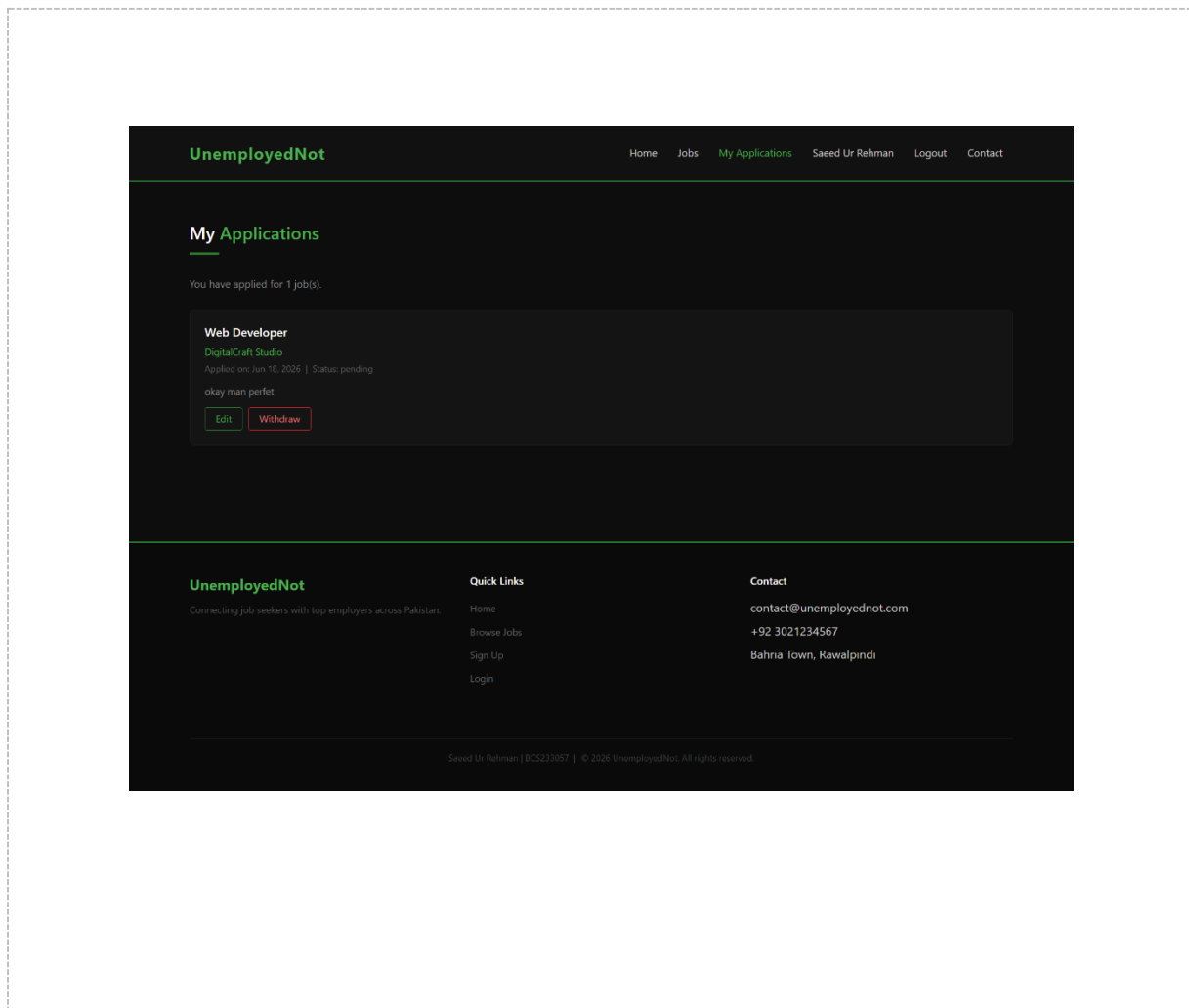
Email: info@unemployednot.com

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Address: 123 Main Street, Karachi, Pakistan

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7.3 My Applications Page



8. Conclusion

UnemployedNot started as a static front-end project and has been successfully converted into a fully dynamic, database-backed web application as required for this assignment. It now supports complete user authentication, full CRUD functionality on job applications, and a smooth, reactive user experience across the frontend.

Along the way, a few real problems came up, a foreign key constraint issue, a navbar that would not update, and a page that kept redirecting unexpectedly, and each one was identified and fixed with a targeted solution rather than a workaround. The result is a working job portal that demonstrates practical integration between an Angular frontend and an ASP.NET Web API backend, connected through a SQLite database.